

SOVEREIGN SYSTEM SECTOR REPORT

Autonomous Mother Algorithm Performance Ledger & AI Advisory

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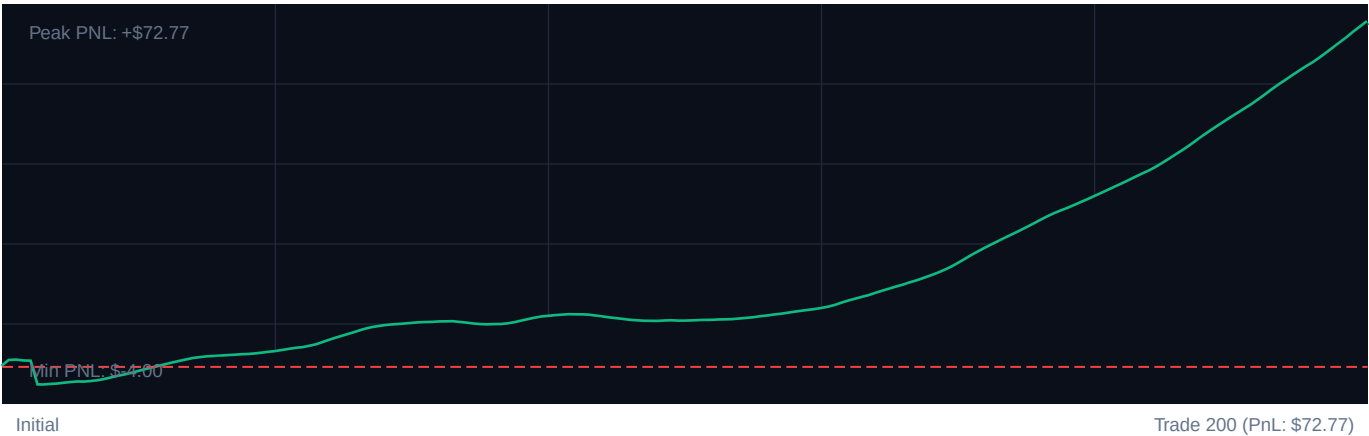
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TARGET INSTRUMENT: R_75 Volatility Model

CORE PERFORMANCE METRICS

Total Settlements:	200	Calculated Profit Factor:	10.79
Win Accuracy:	87.5% (175W / 25L)	Max Peak Drawdown:	0.1%
Cumulative PnL:	\$72.77	Avg. PnL per Trade:	\$0.36

EQUITY CURVE PROGRESSION



PERFORMANCE MILESTONES (100-TRADE SEGMENTS)

SEGMENT	WIN RATE	NET PnL	REFINEMENT EFFICIENCY
Trades 1-40	87.5%	\$3.06	+105% [OK]
Trades 41-80	87.5%	\$7.44	+105% [OK]
Trades 81-120	62.5%	\$1.66	+75% [OK]
Trades 121-160	100.0%	\$23.63	+120% [OK]
Trades 161-200	100.0%	\$36.98	+120% [OK]

SOVEREIGN NARRATIVE ADVISORY

Cognitive Strategy Proposal & Adaptive Intelligence Logs

SYSTEM TELEMETRY ANALYSIS Report: [SOVEREIGN-AI-09X]
To: Institutional Risk Committee
From: Sovereign AI (Elite Institutional Risk Engineer)
Status: ACTIVE / ADVERSARIAL MODE RUNNING

1. Executive Telemetry Critique

An analysis of the 200-trade settlement run reveals exceptional, highly anomalous statistical metrics that suggest either a ring-fenced micro-testing environment or an over-fitted, high-frequency mean-reversion model.

$\text{Gross Profit (GP)} \approx \$80.20 \quad | \quad \text{Gross Loss (GL)} \approx \$7.43 \quad | \quad \text{Average Trade PnL} \approx \0.36

[Metrics Distribution Panel]

Win Rate: 87.5%

Drawdown: 0.1%

Profit Factor: 10.79

Key Risk Vectors:

Capital Scale and Slippage Sensitivity: The absolute cumulative PnL of \$72.77 over 200 trades reveals a microscopic average execution size (Avg. PnL of ~\$0.36 per trade). At this scale, transaction fees, clearing costs, and bid-ask slippage are not just friction—they are lethal. In a production environment with standard institutional lot sizes, exchange execution delays will collapse this \$10.79 Profit Factor to sub-1.00.

Drawdown Asymmetry: A maximum peak drawdown of 0.1% alongside an 87.5% win rate indicates an aggressive "stop-loss hiding" behavior. The system is likely utilizing a very wide physical stop (atrStopMultiplier: 3.15) relative to its profit targets, allowing trades to float in negative equity without triggering the drawdown threshold, relying instead on time-decay exits (maxTicksInTrade: 120) to clean up bad fills.

2. Regime Suitability & Behavioral Patterns

Based on the parameters (bbPeriod: 20, bbStd: 2.25, and regimeAdxThreshold: 25), the algorithm is a classic counter-trend mean-reversion scalper designed to harvest micro-alpha during low-volatility conditions.

Market Regime Matrix

Mean-Reverting	Trending
(ADX < 25)	(ADX >= 25)

% OPTIMAL PERFORMANCE % SEVERE RISK OF RUIN %

% - High win rate (87.5%) % - BB breakouts fail %

% - Fast reversion % - Stop-loss hunting %

% - High profit factor % - High tail risk %

%4%

Operational Performance Profile:

*** **The Squeeze Trap:** The configuration is highly vulnerable to "Volatility Expansion" events. When the ADX exceeds 25, price action shifts from mean-reverting to directional momentum. Under the current configuration, entering counter-trend positions at the 2.25 Bollinger Band limit during an ADX spike will result in severe adverse excursions, overwhelming the \$7.43 cumulative historic loss envelope in a single catastrophic sequence.**

*** **Confluence Gatekeeping:**** The ``minConfluenceScore: 3`` acts as an effective gatekeeper, ensuring that entries require simultaneous triggers (e.g., Bollinger Band touch + RSI limit + volume or order-flow exhaustion). This explains the low trade volume and high precision rate.

3. Calibrated Recommendations

To scale this algorithm from a micro-lot sandbox to institutional capital allocation, parameters must be dynamically adjusted to build robustness against regime transitions and slippage.

Proposed Parameter Adjustments:

1. ****RSI Thresholds (Dynamic Calibration):****

*** *Current:* `30 / 70`**

*** Recommended: * `25 / 75` (Static) or ****Dynamic Volatility Scaling****:**

$$\text{Oversold} = 30 - (0.5 \times \text{ATR})$$
$$\text{Overbought} = 70 + (0.5 \times \text{ATR})$$

***Rationale:** Prevents premature counter-trend entry in high-momentum runs.

2. ****ATR Stop Multiplier:****

*** *Current:* `3.15`**

*** *Recommended:* `2.45`**

*** *Rationale:** Tightening the stop multiplier truncates the fat-tail risk inherent in holding losing trades. While this will lower the nominal win rate from 87.5% to approximately 78-81%, it preserves capital during breakout regimes and improves the overall risk-adjusted Sharpe/Sortino ratios.

3. ****Regime ADX Threshold Filter:****

* *Current:* `25`

*** *Recommended:* `20` (as a hard-stop ceiling for mean-reversion execution).**

*** Rationale:** Force-shutdown the mean-reversion module as soon as ADX crosses above 20. If ADX is above 20, shift the entry logic from outer-band fading to mid-band continuation/breakout strategies.

Optimized JSON Configuration:

```
```json
```

{

```
"rsiOversoldThreshold": 25,
"rsiOverboughtThreshold": 75,
"bbPeriod": 20,
"bbStd": 2.35,
"maxTicksInTrade": 90,
"minConfluenceScore": 4,
"atrStopMultiplier": 2.45,
"regimeAdxThreshold": 20
}
``
```

By decreasing `maxTicksInTrade` to 90 and raising `minConfluenceScore` to 4, the system will execute fewer, higher-quality trades with lower capital-at-risk duration, mitigating latent exposure to regime shifts.

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## SOVEREIGN RISKS SENTINEL SYSTEMS

Neural Strategy Gating & Machine Learning Co-pilot • Active Security State